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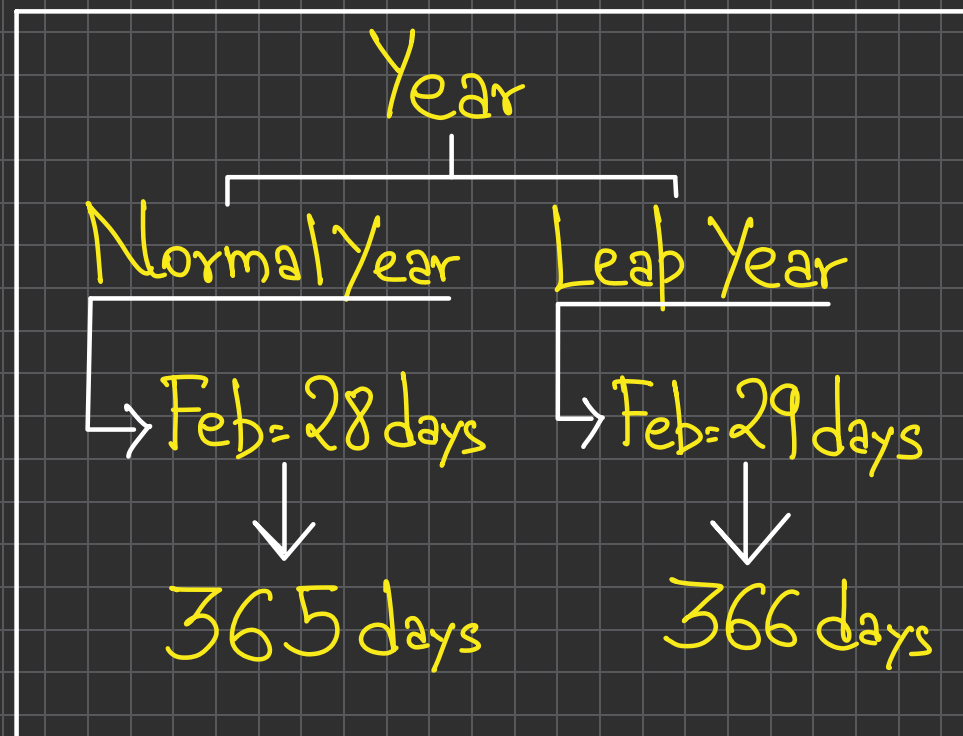
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- ↳ Century year

- A Month Calendar

January - 31 days
February - 28/29 days
March - 31 days
April - 30 days
May - 31 days
June - 30 days
July - 31 days
August - 31 days
September - 30 days
October - 31 days
November - 30 days
December - 31 days



* If last 2 digit of the year is divisible by 4, then it is a **leap year**.
→ 2024 $\rightarrow$ Divisible by 4 = Leap year

* But in a the century, only those century are leap year which are completely divisible by 400.
 $\rightarrow$ 2000 $\rightarrow$ leap year
 $\rightarrow$ 1900 $\rightarrow$ NOT a leap year

Leap year $\rightarrow$ Not a century year $\rightarrow$ Last 2 digits divisible by 4
 $\rightarrow$ Century year $\rightarrow$ The year divisible by 400

Q) A person's date of birth is 29th Feb, 1896. A) 1900 B) 1897
When will he celebrate his next birthday. C) 1904 D) 2000

29th Feb, it means 1896 is a leap year
so next leap year will be $1896 + 4 = 1900$

But 1900 is a century year & not divisible by 400.
 $\therefore$ Next leap year $1900 + 4 = \underline{1904}$

Option C

Q> Given years 1200, 1300, 1400, 1500, 1600, 1700, 1800, 1900, 2000, 2022, 2028, 2046, 2050, 2052, 2072
 A> 5 B> 6 C> 7 D> 4

Leap year: 1200, 1600, 2000, 2028, 2052, 2072
 ↳ Total: 6
 Option B

Q> 100 years having how many leap year?
 A> 25 B> 24 C> 26 D> 22

$\frac{100}{4} = 25$. But the 100th year is not divisible by 400
 $\therefore 25 - 1 = 24$, Option B

Q> In 400 years how many leap year?
 A> 100 B> 96 C> 97 D> 99

1-100 $\Rightarrow$ 24
 101-200 $\Rightarrow$ 24
 201-300 $\Rightarrow$ 24
 301-400 $\Rightarrow$ 25
 97 $\therefore$ Option C

Note: 100 years: 24
 400 years: 97

Number of Odd days

Those days which does not make a whole week are called odd days

OR

When number of days divided by 7, the remaining days are called odd days

100 days $\rightarrow$ 14 weeks + 2 days $\swarrow$ Odd days
 365 days $\rightarrow$ 1 odd day

$\Rightarrow$ In the normal year,
1st January & 31st December

both have the same day

$\Rightarrow$ While in the leap year,
1st January & 31st December
will have successive days

Q> 1st Jan 2007 $\rightarrow$ Fri
 31st Dec 2009 $\rightarrow$?
 A> Fri B> Sat
 C> Mon D> Tue

2007 $\rightarrow$ Normal year 1st Jan = 31st Dec = Fri
 2008 $\rightarrow$ Leap year $\rightarrow$ 1st Jan = Sat, 31 Dec = Sun
 2009 $\rightarrow$ Normal year
 1st Jan = Mon = 31 Dec $\rightarrow$ Option C

* 31st July 2024 → Wed
31st July 2025 → ?

When the upcoming year is having
① Feb - 28 days → Increase by 1 day
② Feb - 29 days → Increase by 2 days

∴ 31st July 2025

Feb, 2025 → 28 days

∴ 31st July, 2025 → Thursday

Qy 20th Feb, 2011 → Monday
20th Feb, 2012 → ?

Feb, 2012 is a leap year

but the date is 20th Feb, 2012, not exceeding 29th Feb

∴ 20th Feb, 2012 → Tuesday

{ 20th Feb, 2013 → Thursday } ← Crossed the date 29th Feb 2012

To find the day, when a date is given

Number code

• Normal year - 144025036146

Each digit represent each month. 1-J, 4-F, 4-M...

• Leap year - 034025036146

Note: Memorise this code to implement it directly in solving the day based questions

<u>Remainder code (Odd day)</u>		<u>Year Code</u>	Same format
01 - Sun	05 - Thurs	1800 - +2	
02 - Mon	06 - Fri	1900 - 0	
03 - Tue	00 - Sat	2000 - +6	
04 - Wed		2100 - +4	
		2200 - +2	

Step 1 → Day of the Month

Step 2 → Month code (Normal year or Leap Year)

Step 3 → Last 2 digit of the year

Step 4 → Count no. of years divisible by 4 passed (Leap year)

Step 5 → Add year code

Step 6 → Take sum of each step (1-6) and divide by 7
The remainder will determine the day of that date

Q 26 Jan 1950.

50/4 → 12

26 - Day

1 - Month Code (Normal year)

50 - Last 2 digits of Year

12 - No of years divisible by 4 passed

00 - Year code

$$\begin{array}{r} 89 \\ 7 \overline{) 89} \\ \underline{77} \\ 12 \\ 7 \overline{) 12} \\ \underline{7} \\ 5 \end{array} \rightarrow 05 - \text{Thursday}$$

Q 16 Dec 1971

144025036146
↓
Dec

16 - Day

6 - Month Code (Normal year)

71 - Last 2 digits of the year

17 - No of years divisible by 4 passed

00 - Year code

$$\begin{array}{r} 17 \\ 4 \overline{) 71} \\ \underline{4} \\ 31 \end{array}$$

$$\begin{array}{r} 110 \\ 7 \overline{) 110} \\ \underline{70} \\ 40 \\ 7 \overline{) 40} \\ \underline{35} \\ 5 \end{array} \rightarrow 05 \rightarrow \text{Thursday}$$

Q 14th May, 1985

144025036146
↓
May

14 - Day

02 - Month Code (Normal year)

85 - Last 2 digits

21 - Years divisible by 4

00 - Year Code

$$\begin{array}{r} 122 \end{array}$$

$$\begin{array}{r} 17 \\ 7 \overline{) 122} \\ \underline{7} \\ 52 \\ 7 \overline{) 52} \\ \underline{49} \\ 3 \end{array}$$

03 - Tuesday

Q 31-07-2024

034025036146
↓
7

31 - Day

0 - Month Code (Leap year)

24 - Last 2 digits

6 - Years divisible by 4

06 - Year Code

$$\begin{array}{r} 67 \end{array}$$

$$\begin{array}{r} 9 \\ 7 \overline{) 67} \\ \underline{63} \\ 4 \end{array}$$

Wednesday

10 May 1857

144025036146
↓
May

10 - Day

02 - Month Code (Normal year)

57 - Last 2 digits of year

14 - Years divisible by 4

02 - Year Code

$$\begin{array}{r} 85 \end{array}$$

$$\begin{array}{r} 12 \\ 7 \overline{) 85} \\ \underline{7} \\ 15 \\ 7 \overline{) 15} \\ \underline{14} \\ 1 \end{array}$$

Sunday

Q) If on 5th June 1960 was Tuesday.
What was the day on 26 Jan 2001.

Verify the day on 5th June, 1960 valid or not.

05 - Day (034025036146)

05 - Month Code

60 - Last 2 digits of the year

15 - No of years divisible by 4

00 - Year Code

85
7 $\overline{) 12}$
7 $\overline{) 85}$
7
15
14
1
→ Sun → Tue (+2)

Therefore for 26th Jan 2001

26 - Day

01 - Month code

01 - Last 2 digits of the year

00 - No of years divisible by 4

06 - Year code

34
7 $\overline{) 4}$
7 $\overline{) 34}$
28
6
→ Friday → Sunday (+2)

Q) 15th Aug 2002 → Sun
Then 10th Dec 2024 → ?
A) Mon B) Wed
C) Fri D) None

15th Aug 2002 (144025036146)

→ 15
03
02
00
06
26
7 $\overline{) 26}$
21
5
→ Thursday (+3) → Sunday

10th Dec 2024 (034025036146)

10
06
24
06
06
52
7 $\overline{) 52}$
49
3
→ Tue (+3) → Friday
∴ Option C

Q) If in a normal year 20th March $\rightarrow$ Sun
What will be the day on 20th Dec of the same year

As per the month code 144025036146
(Here it does not matter that it is a leap year or normal year as Mar, Dec does not change)

144025036146
↓ ↓
March Dec

If we observe any date on March will have (+2) on the same date of Dec.

$\therefore$ Sun $\xrightarrow{(+2)}$ Tuesday

Q) 15th Jan $\rightarrow$ Fri
15th Nov $\rightarrow$?

144025036146
↓ ↓
Jan Nov

$\therefore$ Fri $\xrightarrow{(+3)}$ Monday

Q) 3rd Feb $\rightarrow$ Sat
15th Aug $\rightarrow$?

144025036146
↓ ↓
Feb Aug

3rd Aug $\rightarrow$ Sat $\xrightarrow{(-1)}$ Friday

$\therefore$ 15th Aug $\rightarrow$ Wednesday

Q) In a leap year, 11 Jan $\rightarrow$ Sun
22 Oct $\rightarrow$?

034025036146
↓ ↓
Jan Oct

$\therefore$ 11 Oct $\rightarrow$ Sun $\xrightarrow{(+1)}$ Monday

22 Oct $\rightarrow$ Friday

Similar Month Calendar

In a normal year, which 2 month have Same Calendar

1 4 4 0 2 5 0 3 6 1 4 6

→ Jan(01) = Oct(01)

→ Feb(04) = Mar(04) = Nov(04)

→ April(00) = July(00)

→ Sept(06) = Dec(06)

Note: May(02), June(05) & Aug(03)
are always unique

In Leap year

0 3 4 0 2 5 0 3 6 1 4 6

→ Jan(00) = April(00) = July(00)

→ Feb(03) = Aug(03)

→ March(04) = Nov(04)

→ Sept(06) = Dec(06)

Note: Unique Months - May(02), June(05), Oct(01)

Similar Year Calendar

In such type of Question, last 2 digit of the year divide by 4, find out the remainder

Remainder	Repeat
1	→ +6
2,3	→ +11
0	→ +28

• → 2025
 ↳ 1 → +6 → 2031

• → 2024
 ↳ 0 → +28 → 2052

Note: Every calendar repeats after 28 years
even if normal year or leap year but to find the
nearest repeat year we use the remainder of 4 to get

Ex: 1983 → 2011 (28 years)

 ↳ 1994 (11 years) ← nearest year

If it is a century year.

→ 2000 → $\frac{2000}{4} = 28 \Rightarrow 2000 + 28 = 2028$.
 (Note: 2000 is a Leap year)

→ 1900, century year but not a leap year

Then count number of odd days every year

Year	Odd days
1900	1
1901	1
1902	1
1903	1
1904	2 (Leap year)
1905	1
1906	1
1907	1
1908	2 (Leap year)
1909	1
1910	1
1911	1

→ Total is 7
 so 1906 will be
 same as 1900

→ Total is 7
 Again 1912 will
 be same as 1900
 but it is a LEAP YEAR

A month Calendar

Let's assume, 1st is Sunday it is a month of 31 days

Sun	1	8	15	22	29
Mon	2	9	16	23	30
Tue	3	10	17	24	31
Wed	4	11	18	25	
Thu	5	12	19	26	
Fri	6	13	20	27	
Sat	7	14	21	28	

5 times occurrence

4 times occurrence

If we observe one thing
 the month having 31 days, the days fall on 1, 2, 3
 will occur 5 times in that month

Here Sun, Mon, Tue

- Similarly for 30 days → 1, 2 will have the same day 5 times in that month.
- Feb → 29 days — 1st will have 5 times occurrence
 ↳ 28 days — No 5 times occurrence

Q In a month, how many minimum working days if Sunday is a holiday & 2nd, 4th Sat is a holiday

Minimum days means a month of minimum days.
Feb with 28 days

→ 4 Sunday → 4 holiday — 4
→ 4 Saturday → 2nd, 4th holiday — 2

∴ No of working days: $28 - 6 = 22$

Q Maximum working days, Sunday, 2nd & 4th Sat holiday

Max days = Month of max days = 31 days

Where 4 Sundays + (2nd & 4th Sat) = $4 + 2 = 6$

∴ No of working days = $31 - 6 = 25$

Q In June 2009, determine the first Sunday date

1st June 2009

144025036146
June

01 - Day

05 - Month code

09 - Last 2 digits of the year

02 - No of leap year passed

06 - Year code

23

$$\begin{array}{r} 3 \\ 7 \overline{) 23} \\ \underline{21} \\ 2 \end{array} \rightarrow \text{Monday}$$

1st June → Monday... 7th June is Sunday

Q If today is Friday, what will be the day after 59 days.

Today is Friday, find the odd days

$$\begin{array}{r} 8 \\ 7 \overline{) 59} \\ \underline{56} \\ 3 \end{array}$$

3 → Friday + 3 = Monday

Q If today is Friday, find the day on 59th day

Today is Friday,

$$\begin{array}{r} 8 \\ 7 \overline{) 59} \\ \underline{56} \\ 3 \end{array}$$

Friday + (3-1) = Sunday

Note: After 'x' days = Day + odd days
On 'x' day = Day + (odd-1) days

Q If today is Saturday, after 27 weeks 61 days

27 weeks will be cyclic and on the last day of that day will be Saturday again

Saturday + 61 =

$$\begin{array}{r} 8 \\ 7 \overline{) 61} \\ \underline{56} \\ 5 \end{array}$$

Saturday + 5 = Thursday

Q If today is Monday, what will be it on 200th day.

200th day

$$\begin{array}{r} 28 \\ 7 \overline{) 200} \\ \underline{14} \\ 60 \\ \underline{56} \\ 4 \end{array}$$

Monday + (4-1) = Monday + 3
= Thursday

